

JEE MAIN REVISION MASTER: TRIGONOMETRIC EQUATIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	General & Principal Solutions of Trigonometric Equations	3 PYQs	Trigonometric Equations

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
General & Principal Solutions of Trigonometric Equations	$\sin\theta = 0 \implies \theta = n\pi, \quad \cos\theta = 0 \implies \theta = (2n+1)\frac{\pi}{2}, \quad -\sqrt{a^2+b^2} \leq a\cos x + b\sin x \leq \sqrt{a^2+b^2}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 General & Principal Solutions of Trigonometric Equations (Moderate Yield — 3 PYQs)

Core Principle: General solutions: $\sin\theta = \sin\alpha \implies \theta = n\pi + (-1)^n\alpha$. $\cos\theta = \cos\alpha \implies \theta = 2n\pi \pm \alpha$. $\tan\theta = \tan\alpha \implies \theta = n\pi + \alpha$. For equation $a\cos x + b\sin x = c$, real solutions exist if and only if $|c| \leq \sqrt{a^2 + b^2}$.

■ **Caution / JEE Pitfall:** Squaring both sides of a trigonometric equation introduces extraneous roots; always verify candidate solutions back in the original equation.

■ **Speed Trick:** To find the number of solutions in $[0, 2\pi]$, sketch graphs of LHS and RHS and count intersection points.

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): "If $\cos x + \cos^2 x = 1$, then the value of $\sin^2 x + \sin^4 x$ is..." [Ans: N/A]
- [2022 Shift] Q#2 (MCQ): "The trigonometric function $y = \tan x$ in the II quadrant..." [Ans: N/A]
- [2019 Shift] Q#3 (MCQ): "If $\cos x = |\sin x|$ then, the general solution is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2025 (Q#1)

If $\cos x + \cos^2 x = 1$, then the value of $\sin^2 x + \sin^4 x$ is

- (A) -1
- (B) 1
- (C) 0
- (D) 2

Correct Answer: None

Solution Analysis: $\cos x + \cos^2 x = 1 \Rightarrow 1 - \cos^2 x = \cos x \Rightarrow \sin^2 x = \cos x \Rightarrow \sin^2 x + \sin^4 x \Rightarrow \cos x + (\cos x)^2 = 1$